

Consideration of design viability

The proposed design was carefully reviewed based on several relevant extra-functional considerations; a judgment about design viability based on those considerations — the capacity of the proposed solution to address the problem — is clearly realistic and well supported with credible evidence.

Looking at the Animal Lift Frame Design

When building something for real-world use, especially something that helps lift large animals, there's more to think about than just whether it works. We also have to think about things like safety, cost, ease of use, how long it will last, and how people will take care of it later on. Our group worked hard to think about all of these things while building our metal frame animal lift. Our design is safe, affordable, strong, and a realistic way to solve the problem of lifting heavy animals.

Cost and Budget

Our budget for this project was \$250. That might not sound like a lot when you're trying to build something that can lift an animal weighing up to a ton, but we made it work by being smart with how we spent our money. We had leftover steel from the previous groups allowing us to save our budget on that and spend it on other things like the chain, engine levelers and second track. We also planned ahead and made sure we didn't waste any materials. All the parts we bought had a specific purpose, and we avoided buying anything we didn't need.

The engine levelers were a big part of our design. We used them to help balance the animal during lifting. These tools were much cheaper than buying a more complex lifting system, and they did exactly what we needed them to do. By using our money carefully, we stayed under budget and still built something that works well and is safe to use.

Safety

Safety was one of the most important things we thought about while designing and building the lift. The frame needs to be safe not just for the animal being lifted, but also for the people standing nearby. We tested the frame by hanging a 1-ton weight from it for several minutes. It held the weight without bending or breaking. Nothing moved, and there were no signs that the frame was under too much stress.

We also made sure all the welding was strong. Welds are the places where the metal pieces are joined together, and if they aren't done well, the whole frame could break. We took our time to make sure each weld was clean and solid. That way, the frame could hold a heavy load without coming apart. We believe that our testing and strong welding show that the frame is safe for real use.



<https://www.istockphoto.com/photo/a-strong-man-welder-gm1026970050-275365395>

Strength and Durability

The animal lift will be used indoors, so we don't need to worry about things like rain or wind. Still, we wanted the frame to last a long time. Steel was a good choice because it doesn't wear down quickly and can handle a lot of weight over and over again. The way we built the frame, with lots of support bars and bracing, also helps it stay strong for a long time.

By spreading out the weight and giving the frame good support from all sides, we made sure that no one part of the frame would wear out faster than the others. Even though the frame won't be outside, it's still strong enough to last for many years with regular use.

Ease of Use

Our design needs about 3 to 4 people to operate safely. This is because the frame is large and heavy, and the two engine levelers need to be controlled at the same time. One person alone couldn't lift the frame or guide the animal safely while using the levelers. But once the lift is set up, it's easy to use. The cable system the zoo has allows the team to slowly raise or lower the animal while keeping it balanced.

There are no complicated machines or buttons, just simple tools that are easy to learn. Zoo staff or other users could be trained quickly. Even though the lift takes a team to use, it's still realistic because most animal lifting jobs require more than one person anyway.



As seen in this picture there will be plenty of people in the zoo able to help attach and detach the frame.

Animal Welfare: Keeping the Animal Safe and Calm

One of the most important parts of our project was making sure that the animal would be safe and comfortable during the lift. The frame is strong and steady, which means

the animal won't shake or sway while being lifted. This is very important because animals might get scared or hurt if the lift moves around too much.

The engine levelers help keep the animal balanced, so it doesn't tilt to one side. We also made sure to get rid of any sharp edges or open gaps in the frame, so the animal doesn't get scratched or stuck. Everything about the lift was designed to be gentle and secure for the animal. We believe that this shows we care not just about how the lift works, but also about the animal's well-being.

Easy to Check and Fix

The frame is made of steel and has few moving parts, so it won't need much maintenance. Welds can be checked by looking at them to make sure there are no cracks. If one of the engine levelers breaks, it can be replaced easily since it's a common tool. We also told the zoo that if they ever need help fixing the frame, they can email us. We aren't a repair company, but we're happy to help however we can.

Finally, we believe that our design is realistic and ready for real-world use. We stayed under budget, built something strong, and tested it to make sure it could handle the weight. We also listened to experts and got advice that helped make our project better. This lift isn't just a school project—it's something that can actually help people and animals. That's what makes us proud of what we built.